# PROBLEM STATEMENT-3

## Smart Expense Splitting & Settlement System for Group Payments

**Description**

Group payments are common in everyday situations such as shared travel, restaurant bills, hostel expenses, and event planning. Currently, most people rely on informal methods such as manually calculating expenses, using spreadsheets, or messaging apps to track who owes whom. This often leads to confusion, delayed settlements, and inaccurate calculations.

While digital payment platforms allow quick money transfers, they do not effectively manage **multi-person expense tracking and optimized settlement** when multiple transactions occur among a group.

The challenge is to build a **financial expense management platform** that allows users to create groups, record shared expenses, track balances, and automatically calculate the **minimum number of transactions required to settle all debts** within the group.

The system should allow users to split bills using different methods (equal split, percentage split, custom split) and maintain a transparent ledger of transactions. The platform should also integrate with digital payment systems to allow users to settle balances directly within the application.

The focus of this challenge is on **financial transaction modeling, ledger management, and settlement optimization algorithms** rather than AI.

**User**

- College students sharing hostel or travel expenses
- Friends or families managing shared payments
- Event organizers managing group contributions
- Small teams or communities managing collective budgets

**Impact**

- **Reduces financial confusion** in shared expense scenarios.
- **Minimizes number of settlement transactions**, saving time and effort.
- **Promotes transparency** in group finances through clear transaction ledgers.
- **Encourages responsible financial tracking** for group activities.
- **Improves user experience in digital payments** for shared expenses.

**Tech that can be used**

- Transaction ledger databases (SQL / NoSQL)

- Graph algorithms for settlement optimization
- Payment gateway APIs (UPI / Razorpay / Stripe sandbox)
- Mobile or web application frameworks (Flutter, React, etc.)
- Secure authentication systems
- Backend services (Node.js / Java / Python)